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Research paper

CEHMR: Curriculum learning enhanced hierarchical multi-label classification for medication recommendation

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ABSTRACT

The medication recommendation (MR) or medication combination prediction task aims to predict effective prescriptions given accurate patient representations derived from electronic health records (EHRs), which contributes to improving the quality of clinical decision-making, especially for patients with multi-morbidity. Although in recent years deep learning technology has achieved great success in MR, the performance of current multi-label based MR solutions is unsatisfactory. They mainly focus on improving the patient representation module and modeling the medication label dependencies such as drug–drug interaction (DDI) correlation and co-occurrence relationship. However, the hierarchical dependency among medication labels and diversity of difficulty among MR training examples lack sufficient consideration. In this paper, we propose a framework of Curriculum learning Enhanced Hierarchical multi-label classification for MR (CEHMR). Motivated by the category hierarchy of medications which organizes standard medication codes in a hierarchical structure, we utilize it to provide more trustworthy prior knowledge for modeling label dependency. Specifically, we design a hierarchical multi-label classifier with a learnable gate fusion layer, to simultaneously capture the level-independent (local) and level-dependent (global) hierarchical information in the medication hierarchy. In addition, to overcome the diversity of training example difficulties, and progressively achieve a smoother training process, we introduce a bootstrap-based curriculum learning strategy. Hence, the example difficulty can be measured based on the predictive performance of the MR model, and then all training examples would be retrained from easy to hard under the guidance of a predefined training scheduler. Experiments on the real-world medical MIMIC-III database demonstrate that the proposed framework can achieve state-of-the-art performance compared with seven representative baselines, and extensive ablation studies validate the effectiveness of each component of CEHMR.

1. Introduction

Medication recommendation (MR) aims to recommend effective medication combinations given accurate patient representations derived from electronic health records (EHRs). In the past decades, MR has attracted great research interest in the community of clinical and health informatics, which contributes to improving the quality of clinical decision-making especially for severe patients with multi-morbidity [1]. Recently, massive data from EHR systems have provided opportunities for applying deep learning (DL) technology in MR [2–9]. The workflow of MR is briefly illustrated in Fig. 1.

In nature, MR for patients with multi-morbidity is a multi-label classification task, where multiple labels correspond to the medication combination. Mainstream multi-label-based models can generally be decomposed into a patient representation module and a medication classifier. The patient representation module is responsible for extracting low-dimensional discriminative features correlated to medication labels from longitudinal EHRs, while the medication classifier is adopted to predict multiple medication labels based on these features. Accurate and informative patient representations have been proven crucial to various medical predictive tasks such as readmission prediction [10,11], diagnosis prediction [12,13], and health risk prediction [14,15]. Recent MR methods mainly focus on improving

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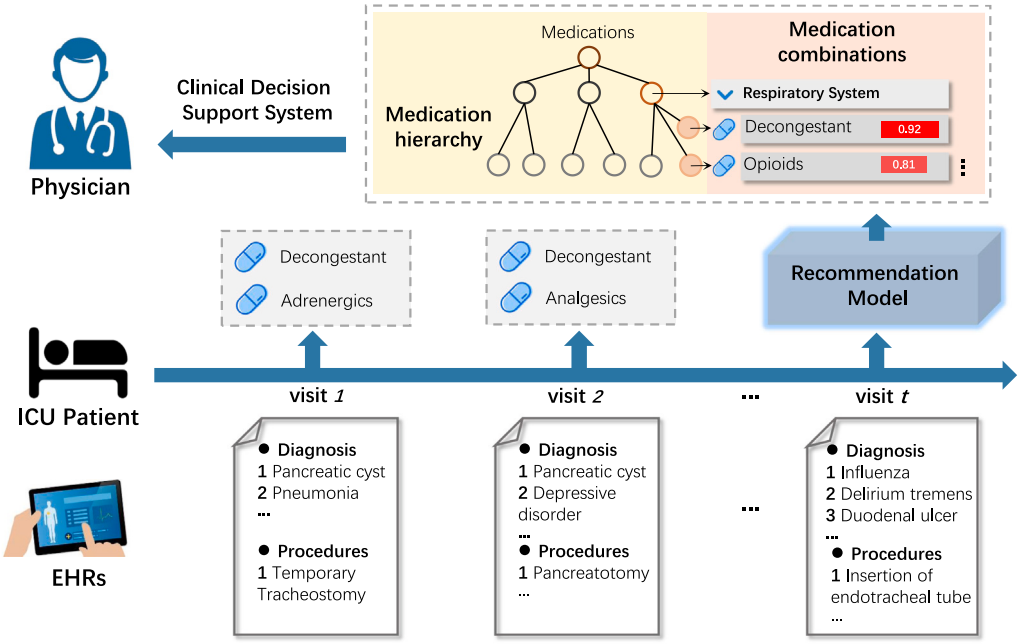

Fig. 1. The main workflow of medication recommendation.

the patient representation module based on dual-view recurrent neural networks (RNNs) [3], transformers [4], graph augmented memory networks [9], etc. They attempt to generate more discriminative features and achieve better prediction performance by explicitly capturing the label/medication dependencies [16,17], such as drug–drug interaction (DDI) correlation and co-occurrence relationship [9].

However, the lack of sufficient consideration for the inherent characteristics of labels and training examples (i.e., the hierarchical dependencies among medication labels and the diversity of difficulty among training examples) should be noted. First, the label hierarchy of medications in which standard medication codes such as anatomical therapeutic chemical codes (ATC,[1] an example is shown in Fig. 2) are organized in a hierarchical structure can provide more trustworthy prior knowledge for modeling label dependency. For instance, given a patient diagnosed with congestive heart failure (ICD-9[2] code: 428.9), when prescribing, an effective reasoning path in accord with clinical expertise may be medication for the cardiovascular system (ATC code: C) → beta-adrenergic blocking agents (ATC code: C07) → beta-adrenergic blocking agents diuretics (ATC code: C07C). Second, there are diverse difficulties among examples for MR model training. The diversity could be three-fold: the disease condition of different patients (especially of patients in ICU) varies greatly; the relations between diseases and medications are quite complicated [6]; the multi-label assumption increases the possibility of partial label missing. The diversity among training data could make the training process less stable and result in unsatisfactory performance. Extensive studies have shown that properly adjusting the usage of training examples can effectively improve performance [18], especially in scenarios where the difficulties of training examples are diverse [19]. Considering these factors, establishing a more efficient and effective MR framework becomes an urgent requirement.

In this paper, we propose a framework of Curriculum learning Enhanced Hierarchical multi-label classification for MR (CEHMR) to tackle the aforementioned challenges. In Fig. 3, we compare the proposed framework with the mainstream multi-label classification framework. Our framework consists of four core components: the patient representation (PR) module, the hierarchical multi-label classifier (HMC),

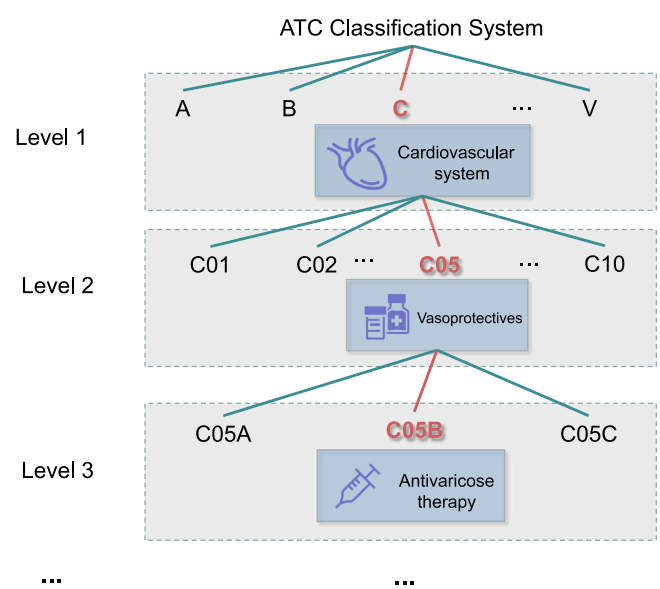

Fig. 2. The main workflow of medication recommendation.

the difficulty measurer (DM), and the training scheduler (TS). PR first derives features from raw EHRs. Then HMC receives the features from PR and outputs the medication combinations (labels of a predefined medication classification system, *e.g.*, ATC). DM and TS work during the training phase to guide the network (PR+HMC) to overcome the diversity of difficulties among training examples. DM ranks training examples by their calculated difficulty metric and TS selects a subset from the whole training set dynamically to optimize model parameters more smoothly.

To implement the CEHMR framework, we utilize a widely used backbone (Retain, [2]) as the PR module to model longitudinal EHRs by reverse time attention layers. To better adapt hierarchical characteristics of medication labels and imitate the decision process of recommending medications, we design the HMC module. HMC is a combination of local and global hierarchical multi-label classifiers (LHC

[1] https://www.whocc.no/atc_ddd_index/.
[2] https://www.cdc.gov/nchs/icd/icd9.htm.

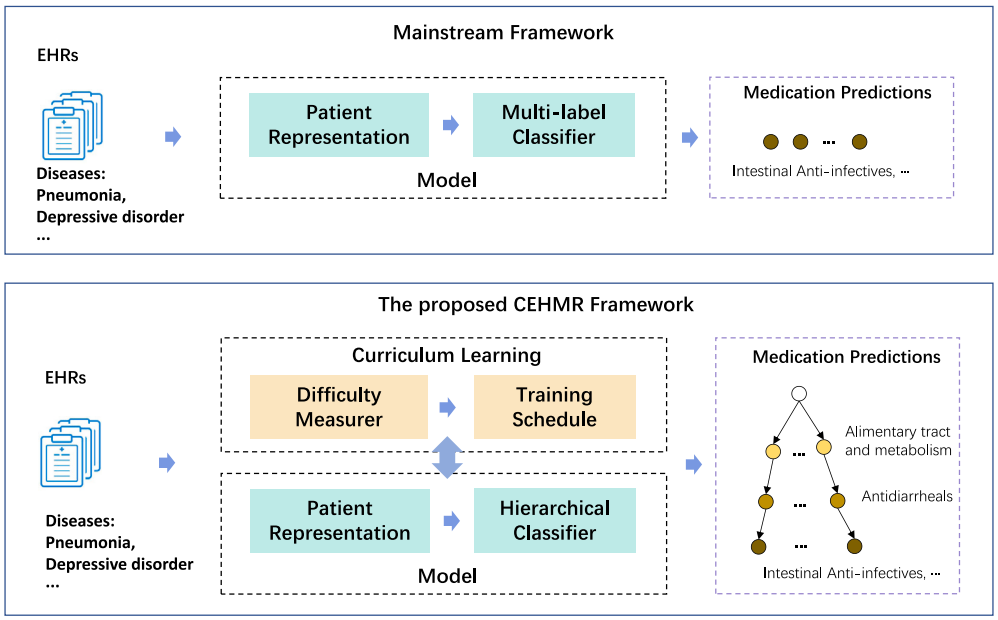

Fig. 3. Comparison between the mainstream multi-label classification based framework and the proposed CEHMR framework.

and GHC, respectively). In specific, LHC predicts the binary probability of each medication label independently while GHC does the same but transfers hierarchical information from top to down by RNNs. This combination can simultaneously capture the level-independent and level-dependent information in the medication classification system. With the combination, GHC can incorporate hidden representations of high-level labels into classification of bottom-level medication labels while LHC can learn independent semantics of each medication label to alleviate error accumulation problem caused by GHC. To encourage the advantages of LHC and GHC, a learnable gate fusion layer is well-designed to fuse the outputs from both classifiers. In addition, to overcome the intrinsic diversity of difficulties among training examples, we propose a bootstrap-based curriculum learning strategy (namely, self-guided curriculum learning (SGCL) algorithm, DM+TS) to train the network. Hence, DM measures the example difficulty based on the predictive performance of a randomly pre-trained network, and then all training examples would be retrained from easy to hard under the guidance of TS. Our experiments show that SGCL can alleviate the bad effects caused by noisy training examples and smooth the training data distribution.

The main novelties of our CEHMR framework can be summarized as (i) CEHMR models the medication label dependency by enabling the classifiers to learn prior hierarchical dependency among medications; The HMC module in CEHMR contributes to classification performance improvement by learning both high-level and bottom-level semantics of medication labels. (ii) With the designed SGCL algorithm, CEHMR can progressively achieve a smoother training process; SGCL algorithm contributes to alleviate effects of noisy training examples and smooth the training data distribution; (iii) In clinical applications, CEHMR can provide more assisted information for clinicians because it predicts both medication combinations and the upper-level medication categories. Experiments on the real-world medical MIMIC-III [20] database demonstrate that the proposed CEHMR framework achieves state-of-the-art performance compared with seven mainstream and representative baselines. Furthermore, extensive ablation studies are conducted, and the results validate the effectiveness of four core components in the CEHMR framework. Our source code is available at GitHub.[3]

[3] https://github.com/siomkos2020/CEHMR.

2. Related work

2.1. Medication recommendation

Deep learning technology in MR mainly includes instance-based methods, longitudinal-based methods, and multi-view sequential learning based methods. Instance-based methods rely on the current encounter to perform recommendations and ignore the longitudinal patient history. Zhang et al. [6] formulate a multi-instance multi-label learning framework and propose a sequence-to-sequence decision model based on a content-attention mechanism to predict the combination of medications. To further utilize the correlation and adverse interactions among medication labels, Wang et al. [7] incorporate DDI knowledge by graph convolutional networks (GCNs) into their framework. Nevertheless, the two methods both lack consideration for valuable longitudinal health information of patients and face performance bottlenecks. As historical medical records can reveal temporal dependencies, more researchers concentrate on longitudinal-based methods in recent years. Shang et al. [3] employ historical diagnosis codes and procedure codes in their patient representation module and then use a graph augmented memory network to query the final medication combinations based on the representation. Similar to [3], Wang et al. [9] adopt the temporal attention mechanism to model the longitudinal EHRs and learn the co-occurrence relationship and DDI correlation with a vanilla GCN. Moreover, Shang *et al.* [4] consider the hierarchical representation of diagnosis and procedure codes and adopt a transformer based network to extract temporal patient representation. Note the asynchronous multi-view sequences of medical events, multi-view sequential learning based methods aim to model intra-view and inter-view interactions. For example, Le et al. [8] utilize dual-view dynamic memory networks to model diagnosis and procedure events. He et al. [5] model correlations between dual-view sequences by self-attention mechanism. It can be found that mainstream MR methods mainly focus on improving the patient representation module and modeling the medication label dependencies like DDI correlation and the co-occurrence relationship. However, they lack sufficient consideration for the inherent hierarchical dependency among medication labels.

2.2. Hierarchical multi-label classification

Hierarchical multi-label classification is a popular subtask in the community of multi-label classification where a hierarchical classifier

Table 1
The notations used throughout the paper.

Symbol	Definition	Symbol	Definition
G_m	The medication label tree	$\boldsymbol{g}_t^*, \boldsymbol{h}_t^*$	The tth hidden layer of GRU_α^* and GRU_β^*
$\mathcal{V}$	The medication node set	s_g, s_h	The dimension of hidden layer in GRU_α^* and GRU_β^*
$\mathcal{V}_l$	The medication node set at level l	$\boldsymbol{w}_\alpha^*, b_\alpha^*$	Weight parameters and bias offset of visit-level attention
$\mathcal{E}$	The edge set that represents the relation of membership	$\boldsymbol{W}_\beta^*, \boldsymbol{b}_\beta^*$	Weight parameters and bias offset of variable-level attention
H	The depth of a given medication label tree	$\tilde{\boldsymbol{q}}_T^*$	The context vector of medical codes $*$
$\boldsymbol{X}$	Longitudinal EHRs of a patient	$\boldsymbol{q}_T$	The patient representation
$\boldsymbol{x}_t^{(n)}$	Multi-variate observations of nth patient at tth visit	s_q	The dimension of hidden layer of MLP_l in LHC
N	The total number of patients	$\boldsymbol{u}_{l,T}$	The GRU hidden layer at lth level in GHC
$T^{(n)}$	The number of visits of the nth patient	$\boldsymbol{W}_l, \boldsymbol{b}_l$	Weight parameters and the bias offset in GHC
T	The number of visits	s_u	The dimension of hidden layer of GRU in GHC
$\boldsymbol{X}^{(n)}$	Longitudinal EHRs of nth patient	$P_{l,T}^L, P_{l,T}^G$	Output probabilities on level l by LHC and GHC
$\boldsymbol{x}_t$	The multi-variate observation at tth visit	P_T^L, P_T^G	Output probabilities by LHC and GHC
$\boldsymbol{c}_t^d, \boldsymbol{c}_t^p, \boldsymbol{c}_t^m$	Diagnosis, procedure and medication codes at tth visit	$\boldsymbol{\gamma}$	Parameters in gate fusion layer
$\boldsymbol{c}_t^*$	The unified notation for different type of medical codes at tth visit	P_T	Output probabilities of HMC
C^*	The unified notation for different type of medical code set	$\mathcal{L}_c, \mathcal{L}_m, \mathcal{L}_r$	The BCE loss, MML and hierarchical violation loss
$\boldsymbol{y}$	The multi-hot vector of ground-truth labels	λ_1, λ_2	Weights of loss $\mathcal{L}_m$ and $\mathcal{L}_r$
Y	The ground-truth medication label set	$\mathcal{L}$	The total loss function
$\hat{\boldsymbol{y}}$	Predicted medication labels represented as a multi-hot vector	$v(\cdot)$	The scoring function in SGCL
$\hat{Y}$	The predicted medication label set	$\hat{Y}_{\text{pre}}$	The predicted medication label set by HMR_p
$\boldsymbol{e}_t^*$	The embeded vector of medical codes $*$ at tth visit	N^*	The positive integer set
$\boldsymbol{W}_e^*$	The embedding matrix for medical codes $*$ at tth visit	$g(s)$	The training scheduler with respect to training epoch s
D	The embedding dimension	δ_l, ϵ_l	Hyperparameters in linear scheduler
$\alpha_{1:T}^*$	Visit-level attention weights	δ_e, ϵ_e	Hyperparameters in exponential scheduler
$\boldsymbol{\beta}_{1:T}^*$	Variable-level attention weights	ϵ_0, s_0	Hyperparameters in single-step scheduler

predicts several different paths of a class hierarchy. Typical hierarchical multi-label classification problems are text classification [21–24], image annotation [25], and protein function prediction [26–28]. MR can be regarded as a hierarchical multi-label classification task because the medications are also organized as a class hierarchy. Performing such classification requires optimizing a hierarchical multi-label loss function either locally or globally [29]. Specifically, the local optimization attempts to discover semantics that dictates the class relationships in particular regions of the class hierarchy. To generate the final classification, the classifier follows a top-down strategy that combines the local predictions [30], being responsible for the prediction of any particular node [31]. For global optimization, the objects are associated with their corresponding classes in the hierarchy as a whole [32]. Wehrmann et al. [33] propose a hybrid optimization that combines the advantages and simultaneously optimizes both local and global loss functions. To the best of our knowledge, few works introduce the category hierarchy of medications (e.g., the hierarchical structure of ATC codes) for MR. Inspired by [33], we improve the way to weigh local and global classifiers and design a specific hierarchical classifier in MR.

2.3. Curriculum learning

Curriculum learning (CL), which imitates the meaningful learning order in human curricula, is a training strategy that shows great potential for improving machine learning models. In essence, education for human physicians is usually organized as curricula by providing easier medical knowledge or cases in the beginning and gradually presenting more complex cases of diseases and prescriptions. However, most frameworks in MR overlook the diversity of difficulties of training examples. The core components of curriculum learning are the difficulty measurer and the training scheduler [18]. Typical CL algorithms include the predefined CL and the automatic CL. The former measures the example difficulty by human experts and trains the model with a fixed training schedule, which is widely applied in various computer vision [34–36], data mining [37], and natural language process [38–40] scenarios. Despite the simplicity and effectiveness of the predefined CL, one of the essential limitations is that easy examples for humans are not always easy for models since the decision boundaries of models and humans could be different [41]. In contrast, automatic CL algorithms are more dynamically adaptive to the current training status by measuring the example difficulty automatically or adjusting the training process based on the feedback from the current model performance. According to Wang et al. [18], automatic CL algorithms can be summarized as (1) self-paced learning algorithms [42], which make the model itself act as a teacher and measure the difficulty of training examples according to its losses on them; (2) transfer teacher algorithms [43,44], which invite a stronger teacher model to act as the teacher and measure the difficulty of training examples according to the teacher's performance on them; (3) reinforcement learning (RL) teacher algorithms [45,46], which adopt an RL model as the teacher and dynamically select training data based on the feedback from the student. Our SGCL is a kind of bootstrap-based CL algorithm and can be regarded as a special case of transfer teacher algorithm, that is, the model pre-trained with training data plays the role of a strong teacher [44].

3. Definitions and notations

To facilitate understanding, we introduce some necessary definitions and notations and formulate the problem of MR before we describe the proposed framework in detail. The notations used throughout the paper are summarized in Table 1.

Definition 1 (*Medication Label Tree*)**.** The medications are often organized in categories according to their therapeutic, pharmacological, and chemical properties, and the organ or system on which they act. We denote this classification system as a medication label tree: $G_m = \{(\mathcal{V}, \mathcal{E})\}$, where $\mathcal{V} = \bigcup_{l=1}^{H} \mathcal{V}_l$ is the medication label set, H is the depth of G_m and $\mathcal{V}_l$ is the medication label set on level l; $\mathcal{E}$ is the edge set that represents the relation of membership among medications.

Definition 2 (*Patient Records*)**.** In most longitudinal EHRs, patient encounter information can be represented as a sequence of multi-variate observations: $\boldsymbol{X}^{(n)} = [\boldsymbol{x}_1^{(n)}, \boldsymbol{x}_2^{(n)}, \ldots, \boldsymbol{x}_{T^{(n)}}^{(n)}]$, where $n \in \{1, 2, \ldots, N\}$, N is the total number of patients and $T^{(n)}$ is the number of visits of the nth patient. To reduce clutter, we will describe the algorithms for a single patient and drop the superscript (n) whenever it is unambiguous. Given $\boldsymbol{X}$, the tth visit record $\boldsymbol{x}_t = [\boldsymbol{c}_t^d, \boldsymbol{c}_t^p]$ is a concatenation of corresponding diagnosis codes $\boldsymbol{c}_t^d$ and procedure codes $\boldsymbol{c}_t^p$. We regard the corresponding prescriptions as the gold standard (label, $\boldsymbol{y}$), which consists of several medication codes: $\boldsymbol{c}_t^m$. For simplicity, we use $\boldsymbol{c}_t^*$ to

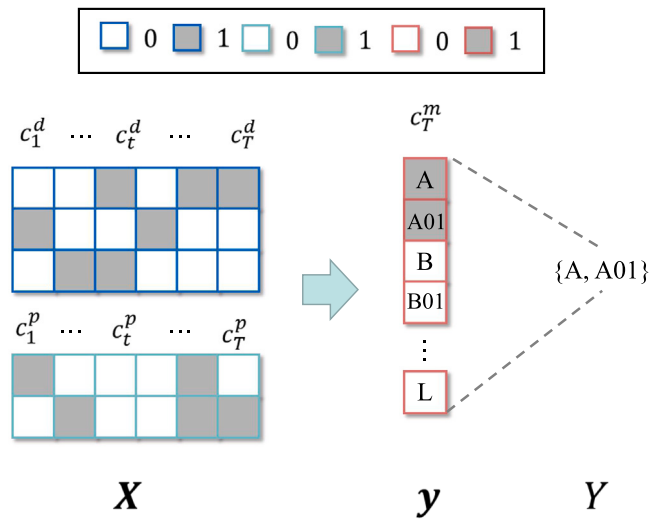

Fig. 4. Illustration of data structure used in this paper. The patient record $\boldsymbol{X}$ contains a temporal sequence of multi-variate observations (c_t^d and c_t^p) where $t = 1 \cdots, T$. Given $\boldsymbol{X}$, the MR model predicts the multi-hot label vector $\boldsymbol{y}$ where its non-zero elements can be converted to a medication set Y. We use squares fulfilled in gray to represent scalar 1 and white to represent scalar 0.

indicate the unified definition for different types of medical codes. $c_t^* \in \{0,1\}^{|C^*|}$ is a multi-hot vector, where C^* denotes the medical code set and $|C^*|$ denotes its size. Given the multi-hot vector $\boldsymbol{y}$, the corresponding multiple medications Y can be directly obtained (as illustrated in Fig. 4).

Problem 1 (*Medication Recommendation*)**.** Given medical records from the patient encounter history $\boldsymbol{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \dots, \boldsymbol{x}_T]$ and the medication label tree G_m, MR requires to predict the medication combination $\hat{Y}$ by generating multi-label outputs $\hat{\boldsymbol{y}} \in \{0,1\}^{|C^m|}$.

4. Proposed framework

In this section, we describe the proposed framework CEHMR in detail. CEHMR is composed of four core components: PR, HMC, DM, and TS. The overall network architecture denoted as HMR (i.e., PR+HMC) is shown in Fig. 5. HMR is trained by the designed SGCL algorithm (DM+TS).

4.1. PR

The PR module is designed to derive discriminative feature vectors from longitudinal EHRs by the reverse time attention mechanism. Our PR refers to a popular medical predictive model [2] as the backbone. Specifically, PR contains the embedding layer, the attention layer, and the representation layer.

Embedding layer. Given $\boldsymbol{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \dots, \boldsymbol{x}_T]$, we first embed c_t^d, c_t^p separately into real vector space as follows:

$$\boldsymbol{e}_t^* = \boldsymbol{W}_e^* \boldsymbol{c}_t^*, \tag{1}$$

where $\boldsymbol{W}_e^* \in \mathbb{R}^{D \times |C^*|}$ is the learnable embedding matrix and the embedding dimension D is a hyperparameter. For simplicity, we denote $\boldsymbol{e}_1^*, \boldsymbol{e}_2^*, \dots, \boldsymbol{e}_T^*$ as $\boldsymbol{e}_{1:T}^*$.

Attention layer. For each medical code sequence, we calculate weights for visit-level attention and variable-level attention respectively. The scalars $\alpha_{1:T}^*$ are the visit-level attention weights that depicts the influences of $\boldsymbol{e}_t^*$ for visit embeddings $\boldsymbol{e}_{1:T}^*$. The vectors $\boldsymbol{\beta}_{1:T}^*$ are the variable-level attention weights that respectively depict the influences of D variables for each embedding in $\boldsymbol{e}_{1:T}^*$. We use two RNNs (specifically, gated recurrent unit, i.e., GRU) to separately generate the attention weights $\alpha_{1:T}^*$ and $\boldsymbol{\beta}_{1:T}^*$ as follows:

$$\begin{aligned}
\boldsymbol{g}_T^*, \boldsymbol{g}_{T-1}^*, \dots, \boldsymbol{g}_1^* &= \mathrm{GRU}_\alpha^* \left(\boldsymbol{e}_T^*, \boldsymbol{e}_{T-1}^*, \dots, \boldsymbol{e}_1^*\right) \\
a_j^* &= \boldsymbol{w}_\alpha^{*\top} \boldsymbol{g}_j^* + b_\alpha^*, \quad j = 1, \dots, T \\
\alpha_1^*, \alpha_2^*, \dots, \alpha_T^* &= \mathrm{Softmax}\left(a_1^*, a_2^*, \dots, a_T^*\right) \\
\boldsymbol{h}_T^*, \boldsymbol{h}_{T-1}^*, \dots, \boldsymbol{h}_1^* &= \mathrm{GRU}_\beta^* \left(\boldsymbol{e}_T^*, \boldsymbol{e}_{T-1}^*, \dots, \boldsymbol{e}_1^*\right) \\
\boldsymbol{\beta}_j^* &= \tanh\left(\boldsymbol{W}_\beta^* \boldsymbol{h}_j^* + \boldsymbol{b}_\beta^*\right), j = 1, \dots, T,
\end{aligned} \tag{2}$$

where $\boldsymbol{g}_t^* \in \mathbb{R}^{s_g}$ is the hidden layer of GRU_α^* at time step t, $\boldsymbol{h}_t^* \in \mathbb{R}^{s_h}$ is the hidden layer of GRU_β^*; $\boldsymbol{w}_\alpha^* \in \mathbb{R}^{s_g}$, $b_\alpha^* \in \mathbb{R}$, $\boldsymbol{W}_\beta^* \in \mathbb{R}^{D \times s_h}$ and $\boldsymbol{b}_\beta^* \in \mathbb{R}^{s_h}$ are the parameters to learn. The hyperparameters s_g and s_h decide the hidden layer dimension of GRU_α^* and GRU_β^* respectively.

Representation layer. Given the attention weights, the context vector $\tilde{\boldsymbol{q}}_T$ is obtained for a patient at Tth visit as follows:

$$\begin{aligned}
\tilde{\boldsymbol{q}}_T^* &= \sum_{j=1}^{T} \alpha_j^* \boldsymbol{\beta}_j^* \odot \boldsymbol{e}_j^* \\
\tilde{\boldsymbol{q}}_T &= \tilde{\boldsymbol{q}}_T^d \oplus \tilde{\boldsymbol{q}}_T^p,
\end{aligned} \tag{3}$$

where $\odot$ means the element-wise multiplication and $\oplus$ denotes the operator of concatenation. In order to enhance the representation capacity of the current visit, we concatenate the context vector with current embeddings, and the final patient representation is:

$$\boldsymbol{q}_T = \tilde{\boldsymbol{q}}_T^d \oplus \tilde{\boldsymbol{q}}_T^p \oplus \boldsymbol{e}_T^d \oplus \boldsymbol{e}_T^p. \tag{4}$$

4.2. HMC

In our framework, HMC is designed for learning the hierarchical structure of the medication label tree from labeled data. HMC is composed of two modules, one of which is a local hierarchical classifier (LHC) and the other is a global hierarchical classifier (GHC). In specific, LHC classifies the input data to each node in the medication label tree while GHC does the same classification but transfers the global information from top to down in the medication hierarchy.

LHC. For a H-level medication label tree, LHC contains H independent multi-label classifiers with respect to each level. Given the patient representation $\boldsymbol{q}_T$, for each level, we use a multi-layer perceptron (MLP) as the multi-label classifier and output the final probabilities over each node on levels as follows:

$$\begin{aligned}
P_{l,T}^L &= \sigma\left[\mathrm{MLP}_l\left(\boldsymbol{q}_T\right)\right] \\
P_T^L &= P_{1,T}^L \oplus P_{l,T}^L \oplus \cdots \oplus P_{H,T}^L,
\end{aligned} \tag{5}$$

where $P_{l,T}^L$ denotes node probabilities on level l, σ denotes the sigmoid function, $\mathrm{MLP}_l(\cdot)$ means a three-layer neural network with a hidden layer dimension of s_q, and P_T^L denotes the output probability of each node in the medication label tree. Note that LHC only captures the local information of each node in the medication label tree but ignores the hierarchical dependencies among medication codes.

GHC. To model the hierarchical dependency among medication codes, GHC utilizes sequential GRUs to transfer global information from top to down in the medication label tree. The GRU hidden layers $\boldsymbol{u}_{l,T} (l = 0, \dots, H)$ are calculated as follows:

$$\begin{aligned}
\boldsymbol{u}_{0,T} &= \boldsymbol{0} \\
\boldsymbol{u}_{l,T} &= \mathrm{GRU}\left(\boldsymbol{q}_T, \boldsymbol{u}_{l-1,T}\right), \quad l > 1,
\end{aligned} \tag{6}$$

where $\boldsymbol{u}_{l,T} \in \mathbb{R}^{s_u}$ and s_u denotes the hidden layer dimension of GRUs in GHC. As GRUs do not exponentially increase the number of parameters with the growth of the medication label tree, GHC will not noticeably increase the computation cost. Note that the way the computation of hidden layers in LHC and GHC are different, where LHC predicts probabilities on level l from $\boldsymbol{q}_T$ while GHC predicts them from $\boldsymbol{q}_T$ and

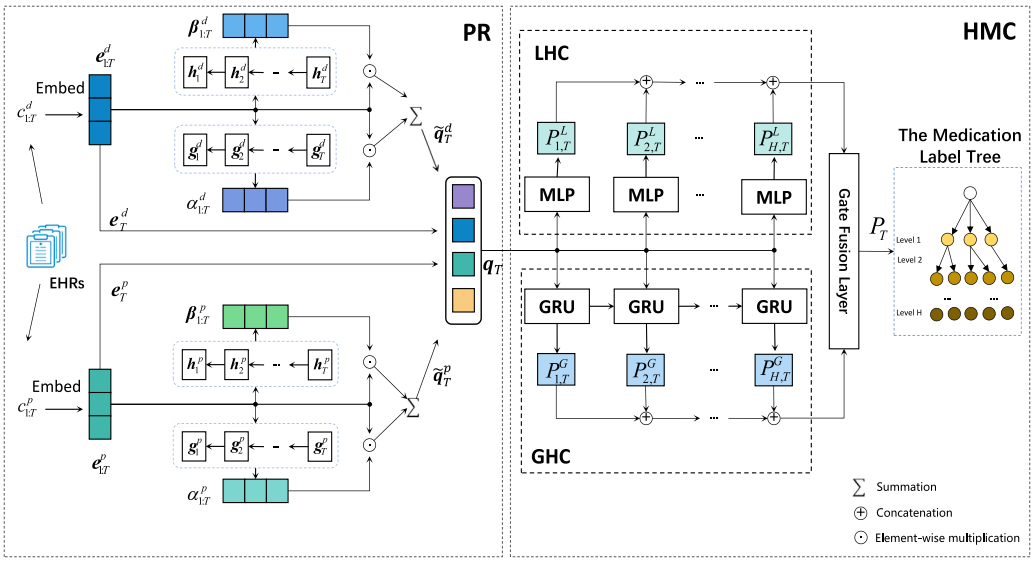

Fig. 5. The overall architecture of the proposed network HMR. HMR is composed of the PR module and HMC. Given EHRs as input, the PR module receives longitudinal medical codes and outputs the feature vector with a fixed dimension for patient representation. HMC classifies the feature vector into multiple node possibilities on the predefined medication label tree.

$l-1$th hidden vector $\boldsymbol{u}_{l-1,T}$. After computing in the GRU hidden layer completed by Eq. (6), we can just use a single-layer perceptron for each level to predict node probabilities and concatenate them together to get global probabilities:

$$\begin{aligned} P^G_{l,T} &= \sigma\left(\boldsymbol{W}_l \mathrm{ReLU}\left(\boldsymbol{u}_{l,T}\right) + \boldsymbol{b}_l\right) \\ P^G_T &= P^G_{1,T} \oplus P^G_{l,T} \oplus \cdots \oplus P^G_{H,T}, \end{aligned} \tag{7}$$

where $\boldsymbol{W}_l \in \mathbb{R}^{|\mathcal{V}_l| \times s_u}$ is the weight matrix $\boldsymbol{b}_l \in \mathbb{R}^{s_u}$ is the bias offset, and $\mathrm{ReLU}(\cdot)$ is the activation function.

Gate fusion layer. Given the outputs from the LHC and GHC, it is intuitive to add them by a fixed scalar weight. However, because of the structural difference between LHC and GHC, they extract information from the observed patient representations from different perspectives. To take the difference between LHC and GHC classifiers on each medication level into account and select the appropriate fusion weight, we replace the scalar weight with a weight vector and add a learnable gate fusion layer with dimension $|\mathcal{V}|$ in our network. Hence, the final output probabilities of HMC are:

$$P_T = \sigma(\boldsymbol{\gamma}) \odot P^L_T + (1 - \sigma(\boldsymbol{\gamma})) \odot P^G_T, \tag{8}$$

where $\boldsymbol{\gamma} \in \mathbb{R}^{|\mathcal{V}|}$ acts as a gate to trade-off the predicted possibilities from LHC and GHC on each node.

Loss function. To optimize the classifiers, we adopt the binary-cross entropy (BCE) loss, the multi-label margin loss (MML), and the hierarchical violation loss. Specifically, the BCE loss enables the network to predict the true label set, which is calculated as follows:

$$\mathcal{L}_c = \sum_{i=1}^{|\mathcal{V}|} \left[y_i \log P_T[i] + (1 - y_i) \log(1 - P_T[i])\right], \tag{9}$$

where $y_i \in \{0, 1\}$ denotes the ground-truth label of ith medication node and $P_T[i]$ means the probability value at ith node at Tth visit. The MML aims to sharpen the output probabilities. MML is defined as:

$$\mathcal{L}_m = \sum_{i=1}^{|\mathcal{V}|} \sum_{j=1}^{|\hat{Y}|} \max\left(0, 1 - \left(P_T[\hat{Y}_j] - P_T[i]\right)\right), \tag{10}$$

where $P_T[\hat{Y}_j]$ means the probability value at jth predicted label at Tth visit. Note that when the generated possibility of node i is larger than the possibility of its parent node $\hat{Y}_{p|i}$, a hierarchical violation happens. Therefore, we employ the hierarchical violation loss $\mathcal{L}_r$ for regularizing hierarchical violations:

$$\mathcal{L}_r = \sum_{i=1}^{|\mathcal{V}|} \left[\max(0, P_T[i] - P_T[p|i])\right]^2. \tag{11}$$

Thus, the final loss function to optimize is given by

$$\mathcal{L} = \mathcal{L}_c + \lambda_1 \mathcal{L}_m + \lambda_2 \mathcal{L}_r, \tag{12}$$

where λ_1, λ_2 are hyperparameters.

4.3. DM

DM acts as a scoring function in curriculum learning, which determines what kind of examples are easier by the specific criteria. In our framework, we regard the network itself as DM and adopt a bootstrap strategy to automatically evaluate the difficulty of a given example. The core idea behind is to pre-train a CEHMR network which is denoted as HMR_p on the whole training set with random sampling order. Then, the difficulty of a given training example $(\boldsymbol{X}, Y)$ corresponding to model predictive ability can be measured. The score $v(.)$ is obtained by calculating the Jaccard similarity between ground-truth labels and predicted labels:

$$v(Y, \hat{Y}_{\mathrm{pre}}) = 1 - \frac{|Y \cap \hat{Y}_{\mathrm{pre}}|}{|Y \cup \hat{Y}_{\mathrm{pre}}|}, \tag{13}$$

where $\hat{Y}_{\mathrm{pre}}$ is the label set predicted by HMR_p.

4.4. TS

Based on the judgment from DM, TS decides the order and size of data subsets throughout the training process. Formally, we denote the TS as $g(s) : N^* \to [0, 1]$, where s is the training epoch, N^* is the positive integer set, and $g(s)$ satisfies $\lim_{s\to\infty} g(s) = 1$. Here, $g(s)$ means the upper bound of difficulty at training epoch s and adjusts the ratio of training examples input into the network dynamically. By properly selecting g, easy examples are trained with high priority and the gradient signal can be smoother. In this paper, we consider three types of mathematical forms for $g(s)$: the linear function, the exponential function, and the step function. In the subsequent section, we will experimentally validate that the exponential scheduler performs best among these schedulers.

Algorithm 1 SGCL algorithm for MR

Input: training set $\{(\boldsymbol{X}, Y)\}$, maximum epoch M, DM $v(\cdot)$, TS $g(s)$;
Output: convergent network HMR*;
Pretrain HMR uniformly on $\{(\boldsymbol{X}, Y)\}$ and obtain HMR_p;
Randomly initialize HMR;
for $s = 1, \cdots, M$ **do**
  for all $(\boldsymbol{X}, Y)$ **do**
    $\hat{Y}_{\text{pre}} \leftarrow \text{HMR}_p(\boldsymbol{X})$;
    Calculate the example difficulty using Eq. 13;
    if $v(Y, \hat{Y}_{\text{pre}}) < g(s)$ **then**
      Calculate loss $\mathcal{L}$ using Eq. 12;
      Update parameters of HMR;
    end if
  end for
end for
return The final network HMR*.

Linear function. The linear scheduler updates the difficulty upper bound linearly at every epoch, which is given by

$$g(s) = \min(\delta_l s + \epsilon_l, 1), \tag{14}$$

where $\delta_l > 0$ is the slope and ϵ_l is the initial difficulty threshold.

Exponential function. The exponential scheduler increases the difficulty upper bound exponentially, which is given by

$$g(s) = \min(e^{\delta_e s + \epsilon_e}, 1), \tag{15}$$

where $\delta_e > 0$ and ϵ_e are hyperparameters.

Step function. The training scheduler with a step function is also called the single-step scheduler, which changes the difficulty threshold once during the whole training process and is given by

$$g(s) = \begin{cases} \epsilon_0, & s < s_0 \\ 1, & s \geq s_0 \end{cases}, \tag{16}$$

where $\epsilon_0 < 1$ denotes the initial difficulty upper bound, and s_0 denotes the step point.

With DM and TS, the pseudocode of the SGCL algorithm is shown in 1. Note that SGCL has good expandability and can be conveniently plugged into other end-to-end deep learning frameworks.

5. Experimental settings

5.1. Data preparation

Data set. We evaluate the CEHMR framework based on a publicly available MIMIC-III database [20] containing the data of patients who stayed in the ICUs at the Beth Israel Deaconess Medical Center. The database includes various kinds of structured encounter information of patients such as diagnosis codes, procedure codes, and medication codes. Among the patients, over 99% of them were diagnosed with multi-morbidity, which satisfies the task scenario of the proposed framework. To transform the raw data into a general format of the data set for machine learning models, we refer to the data preprocessing method similar to Shang et al. [3]. First, the patient records with prescriptions during the first 24-h are selected as this period is often the most critical time for patients to obtain timely and effective treatment. Second, to evaluate the performance of longitudinal EHRs, patients with two or more visits are further selected. Finally, in the MIMIC-III database, diagnoses and procedures are encoded by standard medical coding systems (ICD-9 for diagnoses and ICD-9-CM[4] for procedures), and these codes will be extracted for patient representations. Note that medications in the MIMIC-III database are originally encoded as national drug code (NDC[5]), and in this study are transformed into ATC codes for further acquiring hierarchical information. The DDI knowledge is extracted from the TWOSIDES data set [47], and the top-40 severe DDI types based on [3] are used in our DDI Ratio evaluation. The patients' data are randomly split into train, validation, and test data by a ratio of 4 : 1 : 1. The detailed statistics of the data set are shown in Table 2.

Table 2
Statistics of the data extracted from the MIMIC-III database for MR.

Item	Quantity
The number of patients	6350
The number of clinical events	15 016
The number of diagnosis types	1958
The number of procedure types	1426
The number of medication types	145
The avg number of visits	2.36
The avg number of diagnoses	10.51
The avg number of procedure	8.80
The avg number of medication	3.84
The depth of the medication label tree	3
The code types on first level	14
The code types on second level	72
The code types on third level	145

Medication label tree. To annotate each example with medication labels on high levels (Level 1 to Level H), we extract the medication codes and relations of membership among them from the ATC system and build a tree structure for retrieving. Besides, to reduce the label space of medications and simplify the problem, we only use medication codes from the first 3 levels of the ATC system.

5.2. Evaluation metrics

Considering that the basic framework of MR can be regarded as a multi-label classification task, four metrics are used for evaluation: the Jaccard score, the area under the precision–recall curve (PR-AUC), the F1 score, and the drug–drug interaction rate (DDI Rate). The Jaccard score is calculated as:

$$\text{Jaccard} = \frac{1}{\sum_{n=1}^{N}\sum_{t=1}^{T^{(n)}} 1} \sum_{n=1}^{N}\sum_{t=1}^{T^{(n)}} \frac{|Y_t^{(n)} \cap \hat{Y}_t^{(n)}|}{|Y_t^{(n)} \cup \hat{Y}_t^{(n)}|}, \tag{17}$$

where $Y_t^{(n)}$ and $\hat{Y}_t^{(n)}$ are the ground-truth and predicted medications of the nth patient on the bottom level of the hierarchy respectively.

The averaged metric F1 score can be calculated as:

$$\text{F1} = \frac{1}{\sum_{n=1}^{N}\sum_{t=1}^{T^{(n)}} 1} \sum_{n=1}^{N}\sum_{t=1}^{T^{(n)}} \frac{2 \times \text{P}_t^{(n)} \times \text{R}_t^{(n)}}{\text{P}_t^{(n)} + \text{R}_t^{(n)}}, \tag{18}$$

where $\text{P}_t^{(n)} = |Y_t^{(n)} \cap \hat{Y}_t^{(n)}|/|Y_t^{(n)}|$ and $\text{R}_t^{(n)} = |Y_t^{(n)} \cap \hat{Y}_t^{(n)}|/|\hat{Y}_t^{(n)}|$.

DDI Rate, which reflects the safety of recommended medications, shows the percentage of medication combinations that contain DDIs based on the following formula:

$$\text{DDI Rate} = \frac{\sum_{n=1}^{N}\sum_{t=1}^{T^{(n)}} |\{(c_a, c_b) \in \hat{Y}_t^{(n)} | (c_a, c_b) \in \delta_{ddi}\}|}{\sum_{n=1}^{N}\sum_{t=1}^{T^{(n)}} 1}, \tag{19}$$

where (c_a, c_b) is an interacted medication pair and δ_{ddi} is the set of medication pairs.

[4] https://www.cdc.gov/nchs/icd/icd9cm.htm.

[5] http://www.whocc.no/atc/structure_and_principles.

5.3. Competitors

To evaluate the effectiveness of the proposed framework, we compare it with the following state-of-the-art baselines:

- **Nearest** predicts the medication combinations at tth visit based on previous medications i.e. $\hat{Y}_t = Y_{t-1}$.
- **Logistic Regression (LR)** uses the binary relevance technique and classifies each medication label by a logistic regression model.
- **Leap** [6] is an instance-based model, which uses the attention layer to extract effective information from current patient records and performs recommendation by a sequential decision model.
- **Retain** [2] employs a two-level reverse time attention mechanism to model longitudinal EHRs. It can detect informative past visits and associated clinical variables.
- **DMNC** [8] uses a memory augmented neural network to model the interactions between two asynchronous sequences (diagnosis events and procedure events) and predicts the final medication combinations based on a conventional multi-label classifier.
- **GAMENet** [3] employs a dynamic memory network to store encoded historical medication information, and further utilizes patient representation formed by encoding sequential diagnosis codes and procedure codes to retrieve medications from the memory bank. The retrieved vector is finally concatenated with patient representation to classify the medications.
- **G-BERT** [4] enhances the representation of medical codes based on their hierarchical structures, and integrates the embedding results into a transformer-based encoder for MR. G-BERT performs pretraining on large medical data sets to achieve better performance.
- **AMANet** [5] regards MR as an asynchronous dual-view sequential learning problem. It uses the self-attention layer to capture inter-interactions and intra-interactions in both diagnosis and procedure sequence, and stores encoded historical information in dynamic memory units.
- **TAMSGC** [9] employs the temporal attention mechanism to model the longitudinal EHRs and acquires the co-occurrence relationships and DDI knowledge by a simple graph convolution network. The medication label representation is concatenated with the patient representation to predict the final medication combinations.
- **HMCN** [33] utilizes the multi-hot features as the input and output the hierarchical predictions by combining MLP based classifiers and RNN based hierarchical classifiers.

5.4. Implementation details

In our experiments, the selected values of hyperparameters in CEHMR are listed in Table 3. Among them, we use the same number of units of embedding layers and hidden layers as [3] and search the weights in the loss function from the range $\{0, 0.01, 0.1, 0.5, 1.0\}$ by cross-validation. The network is optimized by an Adam [48] optimizer with a fixed learning rate of 0.0008. To avoid overfitting, we employ the L2-norm regularization technique with a weight decay of 0.001 and the dropout layer after embedding layers with a sampling rate of 0.6. All models are trained for no more than 30 epochs and the parameters that achieve the best performance on validation data during the whole training process are saved. Except for TAMSGC, we use the official public code from GitHub and select the parameters with recommended settings to implement competitors. As the code of TAMSGC is not released, we just report its results in [9] for comparison. We use Python to implement the competitors and our CEHMR on a single NVIDIA V100 device. Whenever our implementation for competitors performs worse than the reported results in the literature, we just compare the reported results instead.

Table 3
Selected values of hyperparameters in CEHMR.

Notation	Description	Value
D	Embedding dimension	64
s_g	The dimension of hidden layer in GRU^*_α	128
s_h	The dimension of hidden layer in GRU^*_β	128
s_q	The dimension of hidden layer of MLP_l in LHC	512
s_u	The dimension of hidden layer of GRU in GHC	256
λ_1	The weight of loss $\mathcal{L}_m$	0.01
λ_2	The weight of loss $\mathcal{L}_r$	0.01

Table 4
Performance (%) compared with state-of-the-art baselines on the MIMIC-III database.

Model	Jaccard	PR-AUC	F1	DDI rate
Nearest	39.11 ± 0.20	38.05 ± 0.17	54.65 ± 0.18	7.91 ± 0.09
LR	40.75 ± 0.21	67.16 ± 0.18	56.58 ± 0.19	7.86 ± 0.09
Leap	38.44 ± 0.24	55.01 ± 0.33	54.10 ± 0.26	$\mathbf{5.32 \pm 0.08}$
Retain	41.46 ± 0.28	66.20 ± 0.33	57.81 ± 0.27	7.97 ± 0.20
DMNC	43.43 ± 0.25	68.56 ± 0.39	59.34 ± 0.30	9.49 ± 0.11
GAMENet	45.09 ± 0.25	69.04 ± 0.30	60.81 ± 0.25	7.49 ± 0.06
AMANet	45.26 ± 0.26	70.13 ± 0.27	61.11 ± 0.27	7.45 ± 0.05
G-BERT	45.65 ± 0.26	69.60 ± 0.26	61.52 ± 0.24	8.04 ± 0.10
TAMSGC	46.61 ± 0.25	$70.50 \pm 0.30^+$	62.25 ± 0.25	7.63 ± 0.07
HMCN	$46.65 \pm 0.33^+$	69.90 ± 0.38	$62.50 \pm 0.24^+$	8.01 ± 0.09
CEHMR(ours)	$\mathbf{47.61 \pm 0.34}$	$\mathbf{71.17 \pm 0.31}$	$\mathbf{63.25 \pm 0.31}$	$7.42 \pm 0.12^+$

*The best result is in bold and the second-best result is labeled by superscript +. The lower the DDI rate, the safer an MR model. Note that the DDI rate on ground-truth medications in test data is 0.0777.

6. Experimental results

6.1. Performance on MIMIC-III database

Table 4 shows the evaluation metrics of the proposed CEHMR compared with state-of-the-art baselines. For fairness, we reproduce the performance of all baselines based on their published code on our data set and select the best results to report. As the paper of G-BERT [4] does not include the DDI Rate, we evaluate and report the DDI Rate of G-BERT on our database. The final row of Table 4 shows the performance of CEHMR and we report the results of the network trained by SGCL with an exponential training scheduler.

In Table 4, Leap [6] achieves the best DDI Rate but sacrifices the Jaccard score. GCN-enhanced models like GAMENet [3] and TAMSGC [9] achieve better performance possibly because of the utilization of the co-occurrence and DDI dependency among medication labels. G-BERT [4], which is similar to GAMENet but enhances patient representations by integrating hierarchical information of medical codes, achieves limited improvement on the Jaccard score. AMANet [5] and DMNC [8] regard MR as a sequential learning problem but their performance is not outstanding compared with GCN-enhanced models possibly due to the evolution of diseases is not stable for patients in ICU. Compared with all baselines, our framework achieves the best Jaccard score, PR-AUC, F1 score, and the second-best DDI Rate. Note that the oracle DDI Rate (calculated based on the ground-truth medications) in test data is 0.0777. The above results can strongly indicate that CEHMR can improve the overall performance as well as keep the safety for MR.

To evaluate the significance of results in Table 4, we apply pairwise t-test regarding our framework CEHMR as the control method to statistically compare the differences between the results over 10 runs of each competitor. The null hypothesis is that there is no significant difference between the results of CEHMR and the competitor. Table 5 shows the statistics of the p-value with respect to Jaccard score, PR-AUC, F1 and DDI-Rate. In Table 5, it is noted that p-values of all competitors are less than 0.05, thus we reject the null hypothesis for any of the evaluation metrics at a significance level of 0.05.

Table 5
Results (p-value) of the pair-wise t-test where our CEHMR framework is the control method.

Model	Jaccard	PR-AUC	F1	DDI-rate
Nearest	3.95e−22	4.86e−34	8.51e−24	2.04e−06
LR	4.26e−21	5.59e−18	1.04e−21	1.17e−04
Leap	6.77e−24	9.94e−28	4.83e−24	5.54e−19
Retain	2.28e−20	9.20e−17	1.56e−19	8.54e−06
DMNC	8.88e−18	4.91e−11	5.99e−14	1.16e−02
GAMENet	1.14e−11	6.75e−10	1.03e−13	1.08e−01
AMANet	1.36e−12	4.59e−07	1.19e−12	1.74e−01
G-BERT	3.67e−11	2.17e−09	4.54e−11	1.65e−07
TAMSGC	6.15e−08	3.87e−06	2.36e−07	1.66e−01
HMCN	3.35e−05	4.31e−06	2.40e−05	3.81e−08

Table 6
Contribution of different components to CEHMR performance.

Model	Jaccard	PR-AUC	F1	DDI rate
CEHMR w/o GHC	46.27 ± 0.28	70.37 ± 0.30	62.02 ± 0.26	7.32 ± 0.07
CEHMR w/o LHC	46.90 ± 0.30	71.09 ± 0.29	62.63 ± 0.29	$\mathbf{7.29 \pm 0.10}$
CEHMR$_{fix}$	46.92 ± 0.33	70.79 ± 0.25	62.63 ± 0.31	8.06 ± 0.11
RandHMR	47.13 ± 0.33	70.90 ± 0.32	62.81 ± 0.23	7.48 ± 0.09
CEHMR	$\mathbf{47.61 \pm 0.34}$	$\mathbf{71.17 \pm 0.31}$	$\mathbf{63.25 \pm 0.31}$	7.42 ± 0.12

6.2. Ablation study

The ablation study is conducted for three targets: (i) to demonstrate the contribution of the proposed HMC; (ii) to demonstrate the effectiveness of the proposed SGCL algorithm; (iii) to demonstrate the contribution of the loss components $\mathcal{L}_m$ and $\mathcal{L}_r$. Table 6 presents the performance results of some CEHMR variants on the MIMIC-III database.

Ablation study of HMC. The medication classifier contains two main submodules: LHC and GHC. LHC is responsible for learning the specific semantics of medication labels and GHC is designed to capture global hierarchical dependencies among medications. We design the following variants of CEHMR to validate the contribution of them:

CEHMR w/o GHC: In this variant, we remove the GHC module from the whole network, so the network cannot transfer the hierarchical information. As a result, the Jaccard score drops drastically to 0.4627. This is because, without GHC, the LHC module only learns the local semantics from the medication hierarchy and can easily lead to overfitting.

CEHMR w/o LHC: In this variant, we remove the LHC module from the whole network. Although the Jaccard score of this variant exceeds CEHMR w/o GHC by about 0.6%, the performance drops a lot compared with the full network. This is because GHC just shares the weight parameters on each level of the medication label tree, ignoring the local label semantics.

CEHMR$_{fix}$: This variant is designed to validate the proposed gate fusion layer. In this variant, we regard γ as a scalar and select it through grid searching, which is similar to [33]. As a result, when $\sigma(\gamma) = 0.4$, the model achieves the best but lower Jaccard score than the full network, which indicates the effectiveness of the gate fusion layer.

Ablation study of SGCL algorithm. The proposed SGCL training algorithm aims to make the network achieve superior performance by adjusting the training process. Here, the following variant is tested to evaluate the contribution of the SGCL algorithm to the overall performance of CEHMR.

RandHMR: In this variant, we sort the training examples in random order and train the network on all examples every epoch. The resulted performance drops by about 0.4%, which indicates the network benefits from a better training algorithm.

Table 7
Contribution of different loss functions.

$\mathcal{L}_c$	$\mathcal{L}_m$	$\mathcal{L}_r$	Jaccard	PR-AUC	F1	DDI rate
✓			45.06 ± 0.32	70.31 ± 0.31	60.80 ± 0.28	$\mathbf{7.18 \pm 0.11}$
✓	✓		47.26 ± 0.33	71.02 ± 0.28	62.96 ± 0.31	7.50 ± 0.10
✓	✓	✓	$\mathbf{47.61 \pm 0.34}$	$\mathbf{71.17 \pm 0.31}$	$\mathbf{63.25 \pm 0.31}$	7.42 ± 0.12

Ablation study of loss function. We are the first to combine the BCE loss, the MML, and the hierarchical violation loss in MR. In Table 7, we validate the effectiveness of the target loss function by gradually adding each kind of loss. It can be found that all loss functions contribute to overall performance improvement.

7. Discussion

7.1. In-depth view of exploring medication hierarchy

To encourage a better knowledge of the medication hierarchy in the design of HMC, three CEHMR variants that use the upper-level medication labels to varying degrees are tested and the results of the Jaccard score are shown in Fig. 6. Among them, L1+L2+L3 is the proposed CEHMR (L1, L2, and L3 mean the first, second, and bottom levels of the medication label tree respectively). In other variants, we remove the loss components related to medication labels on level 1 or level 2 respectively. For example, L1+L3 means in loss $\mathcal{L}$, the components related to medication labels on level 2 are masked by zero. It is common sense that the performance on level 3 should be majorly focused on for MR, and we refer to the performance on level 1 and level 2 for a better understanding of the performance difference. Fig. 6 shows the Jaccard scores in detail on different levels, where interesting phenomena can be found from the results. First, as is shown in Fig. 6(a), the predictive performance drops dramatically as the level grows, which indicates that CEHMR performs better on high-level labels. Second, comparing Fig. 6(a) with (b), it can be found that CEHMR benefits from learning multi-level medication labels and results in an improvement of 0.64% in the Jaccard score on level 3. Predicting upper-level medication labels can be seen as an easier auxiliary task that can regularize the network parameters and increase the model's generalization ability. Third, from Fig. 6(c) and (d), the effect of high-level medication labels differs. The level-2 labels provide more auxiliary and useful information for CEHMR with an improvement of 0.53% in the Jaccard score, because the semantics of medication labels between adjacent levels are closer.

7.2. Case study

To understand how CEHMR works on specific examples, we conduct a case study on test data. We compare the results of CEHMR with its variants: RandHMR, CEHMR no HC, and a competitive baseline G-BERT. Here, CEHMR no HC is the variant that removes the loss components of level 1 and level 2 medication labels. According to the Jaccard score of each encounter record, we randomly select a difficult example (Jaccard score < 0.45) and an easy example (Jaccard score > 0.80), i.e., the record ID (index a specific visit of a patient) is P551V04 and P544V01 respectively. Fig. 7 shows the predicted results of the above models compared with the ground-truth labels. Missed and error medication labels in predictions are marked in red and blue respectively. Based on Fig. 7, we remark some observations in the following.

P551V04 is a difficult example, which can illustrate the effectiveness of HMC. It can be found that for P551V04, all models miss the medication D07A (plain corticosteroids) except CEHMR. Actually, the frequency of occurrence of D07A among all training examples is only 7%, which could make it easily neglected by classifiers. There is a noteworthy phenomenon that CEHMR correctly predicts the branch $D \rightarrow D07 \rightarrow D07A$ (dermatological drugs → corticosteroids, dermatological

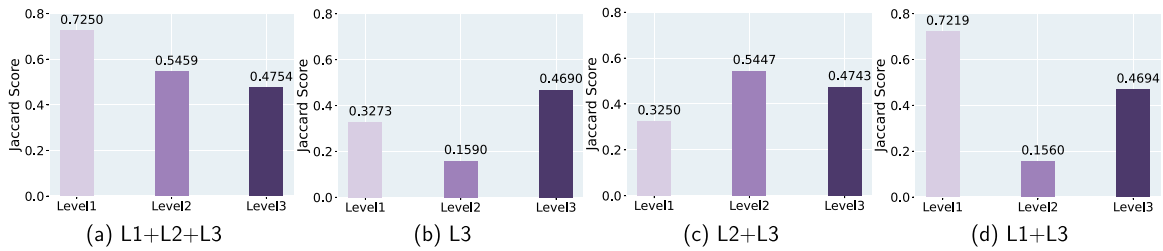

Fig. 6. Jaccard scores of four variants of CEHMR where (a) utilizes all medication labels in the medication label tree, (b) only utilizes the bottom-level (level 3) labels, (c) utilizes labels from level 2 to level 3, and (d) utilizes labels from level 1 and level 3.

Records **Diagnosis codes:** 51884, 5770, 49121, 99731, 5849, 2762, 4871, 0417, 27669, 2809, 2875, 42789, 4019, 7955, V1251, V1582, 30000, V4986, 51889, 7904, 04112, 45829, 4148 **Procedure codes:** 311, 9672, 4311, 3321, 966, 3893	
Ground truth	**L1:** A, B, C, D, N, J; **L2:** A01, A07, A12, B05,C01, D07, N05, J01, J05; **L3:** A01A, A07A, A12C, B05C, C01C, D07A, N05C, J01D, J05A
GBERT	**Predictions:** **L1:-**; **L2: -**; **L3:** A01A, A07A, A12C, B05C, N05C, J01D, N02B, N03A, A02B, C03C, C07A, R01A, R03A, H04A **Comparison:** Missed: **C01C, D07A, J05A;** Errors: **N02B, N03A, A02B, C03C, C07A, R01A, R03A, H04A;**
RandHMR	**Predictions:** **L1:** A, B, C, D, N, J, R; **L2:** A01, A02, A06, A07, A12, B01, B05, C07, C10, N01, N02, N05, J01, R01, R03, H04; **L3:** A01A, A07A, A12C, B05C, N05C, N01A, N02B, A02B, A06A, C07A, C10A **Comparison:** Missed: **C01C, J01D, D07A, J05A;** Errors: **N01A, N02B, A02B, A06A, C07A, C10A;**
CEHMR no HC	**Predictions:** **L1:-**; **L2: -**; **L3**: A01A, A07A, A12C, B05C, N05C, J01D, N01A, N02B, A02B, A06A, C07A, C03C, C10A, R01A, R03A, H04A **Comparison:** Missed: **C01C, D07A, J05A;** Errors: **N01A, N02B, A02B, A06A, C07A, C03C, C10A, R01A, R03A, H04A;**
CEHMR	**Predictions:** **L1:** A, B, C, D, N, J; **L2:** A01,A07,A12, B05,C01, D07, N05,J01, J05; **L3:** A01A, A07A, A12C, B05C, D07A, N05C, J01D, B01A, A07E, B01A, A07E **Comparison:** Missed: **C01C, J05A;** Errors: **B01A, A07E;**

(a)

Records **Diagnosis codes:** 1983, 3485, 1623, 5180, 5990, 78039, V641 **Procedure codes:** 0159, 0212	
Ground truth	**L1:** A, B, C, H, N, J; **L2:** A01, A02, A06, A12, B01, B05, C02, H04, N02, N03, J01; **L3:** A01A, A02B, A06A, A12C, A12B, B01A, B05C, C02D, H04A, N02A, N02B, N03A, J01X;
GBERT	**Predictions:** **L1:- L2: -**; **L3:** A01A, A02B, A06A, A12C, A12B, B01A, B05C, C02D, N02A, N02B, N03A, J01D; **Comparison:** Missed: **H04A, J01X;** Errors: **J01D;**
RandHMR	**Predictions:** **L1:** A, B, C, N, J; **L2:** A01, A02, A04, A06, A12, B01, B05, C02, C07, J01, N02, N03; **L3:** A01A, A02B, A06A, A12C, A12B, B01A, B05C, C02D, N02A, N02B, N03A, C07A, J01D; **Comparison:** Missed: **J01X, H04A;** Errors: **C07A, J01D;**
CEHMR no HC	**Predictions:** **L1:-** **L2: -**; **L3**: A01A, A02B, A06A, A12C, A12B, B01A, B05C, C02D, H04A, N02A, N02B, N03A, C07A, J01D; **Comparison:** Missed: **C01C, D07A, J05A;** Errors: **C07A, J01D;**
CEHMR	**Predictions:** **L1:** A, B, C, D, N, J; **L2:** A01, A02, A04, A06, A12, B01, B05, H04, J01, N02, N03; **L3:** A01A, A02B, A06A, A12C, B01A, B05C, C02D, H04A, N02A, N02B, N03A; **Comparison:** Missed: **A12B, J01X;** No errors;

(b)

Fig. 7. The case study of randomly selected records (ID P551V04 and P544V01) from the MIMIC-III database. The missed medication labels are marked in blue and the errors in predictions are marked in red. (a) A difficult case (ID P551V04) where the disease conditions are more complex (more diagnosis codes and procedure codes). (b) An easier case (ID P54V01) where fewer medical codes are recorded. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

preparations → plain corticosteroids) in the medication label tree. This is because D and D07, the ancestor nodes of D07A, have higher frequencies of occurrence in training data (22% and 8% respectively) and are easier to learn. With the help of high-level auxiliary labels in predictions, our network is more likely to predict labels with a low frequency of occurrence (e.g., D07A) based on the hierarchical dependency among medication labels.

P554V01 is an easy example that all baselines perform well on it. Although the proposed CEHMR misses two bottom-level medication labels, it achieves 100% precision while other models make errors. We attribute this phenomenon to the utilization of the SGCL algorithm. Take the label J01D (other beta-lactam antibacterials like first-generation cephalosporins) for example, all models mispredict it except CEHMR. Note that the ground-truth label J01X (antibacterials like glycopeptide antibacterials or polymyxins) is similar to J01D in usage (they are both antibacterials). Besides, the frequency of occurrence of J01X is much smaller than J01D (about 2% vs. 12% in training data). Therefore, it is challenging to discriminate between J01X and J01D. These facts increase the difficulty of discriminating between them. Comparing CEHMR with RandHMR, CEHMR adds examples gradually

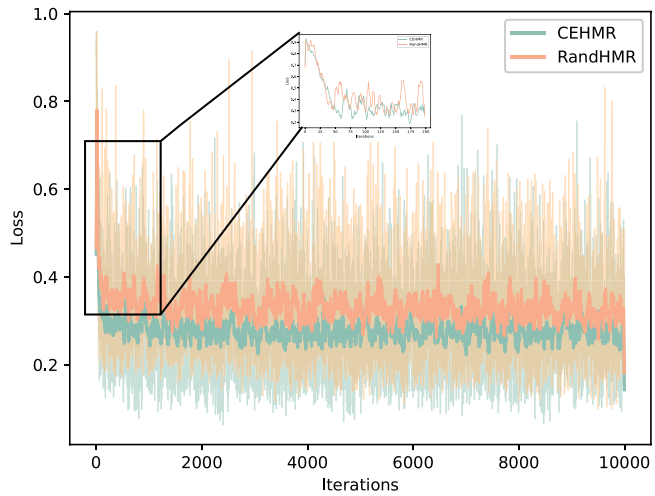

Fig. 8. The loss variation of CEHMR and RandHMR during model training. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

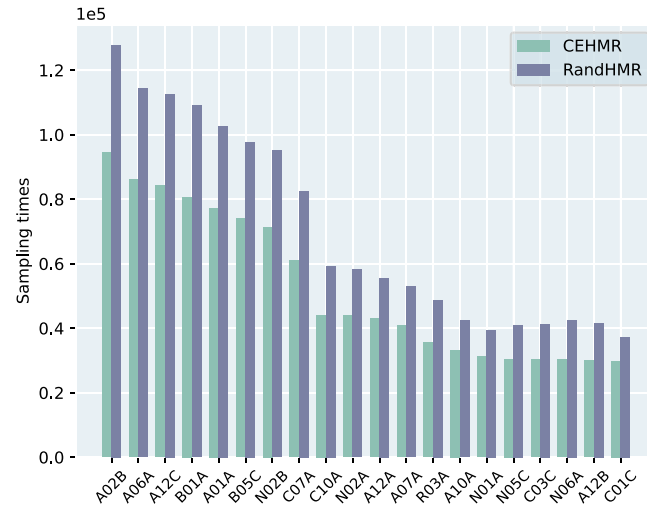

Fig. 9. Top-20 high positive sampling times of bottom-level medication labels in CEHMR and RandHMR.

and achieves a smoother optimization, thus its prediction performs more cautiously and is more likely to avoid errors.

Nevertheless, CEHMR still makes some mistakes, one of the possible reasons is that for MR, structured medical codes are abstract descriptions of patient status while some original but informative features helpful for prescription are overlooked. Symptoms of a patient could be incorporated into our framework to deal with this problem, but we leave it as feature work. In addition, drug consistencies may change over time and across different countries, therefore, the adaptivity to temporal and regional differences is a problem worth investigating.

7.3. Explanation of the effectiveness of SGCL algorithm

Here, we further discuss the effectiveness of the SGCL algorithm from views of the loss optimization, the label distribution affected by SGCL, and the choice of TS.

To explain the effectiveness of the SGCL algorithm, we collect the dynamic loss values of CEHMR and its variant RandHMR respectively. Fig. 8 shows the loss variation of both models during 10000 iterations. In Fig. 8, thin curves depict the actual loss variation, while thick curves adopt mean value smoothing to loss function variation for better visualization. Due to the diversity of difficulties that exists in training examples, the loss functions of both models oscillate intensely and converge slowly (especially shown by the orange curves). After introducing the SGCL algorithm, the loss drops faster and becomes smoother (shown by green curves). Note that at the beginning of training, the loss of CEHMR decreases to a low value faster than RandHMR. This is because easy examples are sampled at first and they constitute a smoother feature space to learn. Due to the flatter target function, the network parameters converge to the local minimum fast and arrive at the global minimum gradually with more difficult examples added. Besides, the SGCL algorithm makes the network trained fewer times on difficult examples, which can alleviate their effects on loss oscillation.

The proposed SGCL algorithm not only affects the optimization of the loss function but also affects the training data distribution by non-uniformly sampling. We can refer to the positive sampling times (the number of occurrences in positive training examples with respect to a specific label) of each bottom-level medication label to compare CEHMR with RandHMR. The results of the top 20 frequent labels are shown in Fig. 9. Obviously, there is a large difference in the distribution of medication labels. On the whole, the medication label distribution of CEHMR is smoother than that of RandHMR, i.e., medication labels with high frequency (such as A02B, A06A, etc.) are sampled fewer times. This change in data sampling enables the network to pay more attention to medication labels with moderate and low frequencies.

Although the SGCL algorithm can help the network achieve a better performance, it depends on the choice of TS that controls the size of the training subset at each epoch. Here, we discuss the effect of different TSs. We refer to Eqs. (14)–(16) and choose the linear function, the exponential function, and the step function as TS, and denote the corresponding variants as CEHMR_{linear}, CEHMR_{exp}, and CEHMR_{step} respectively. We calculate the ratio of used training examples and the Jaccard score on test data at each epoch during training, and the results are shown in Fig. 10. According to Fig. 10.(a), CEHMR_{linear} begins at a low ratio and adds new examples at a high speed, while CEHMR_{exp} begins at a higher ratio and adds new examples more slowly. Different from the continuous schedulers, CEHMR_{step} just divides the training examples into an easy subset and a difficult subset. As shown in Fig. 10.(b), CEHMR_{exp} performs the best among all variants because of the appropriate initial ratio and speed of adding new examples. For CEHMR_{step}, the Jaccard score converges fast due to the initial easy training subset but meets the performance bottleneck when difficult examples are added. Therefore, for the MIMIC-III database, we recommend the exponential function as TS. For other scenarios, automatic selections of TS can be designed with advanced curriculum learning technology, which is worth studying further in the future.

7.4. Limitations

We remark that the HMC module implemented by RNN is specially designed for the label system with tree structure, which is conventional in medical fields. However, for label hierarchy in other HMC tasks, e.g., news text classification, labels are often organized by directed acyclic graphs (DAG). When applying CEHMR to this kind of data sets, some necessary modifications are required. A feasible idea is to convert the DAG into multiple classification paths, realizing the hierarchical classification.

In addition, one possible limitation of the SGCL algorithm could be its heavy dependence on hyperparameters. Only when appropriate hyperparameters (e.g., the training scheduler) are selected, can the optimal performance be obtained, which may lead to extensive manual labor. Moreover, for unbalanced data sets, SGCL intrinsically plays a certain role in reweighting training samples. If SGCL is combined with data sampling or cost-sensitive learning, it could better improve the performance of CEHMR on unbalanced data sets.

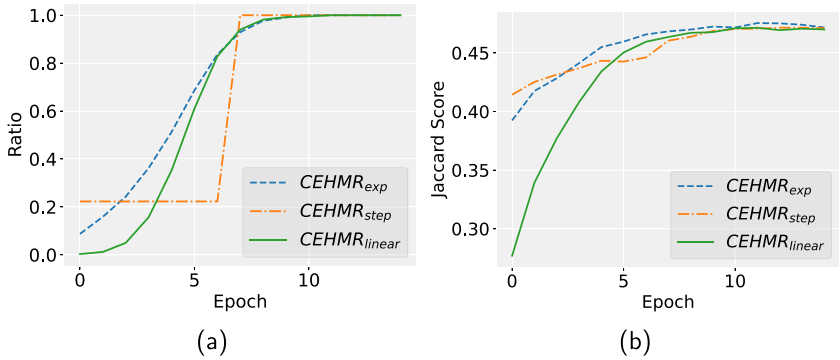

Fig. 10. (a) The ratio of used training examples with different TSs during training. (b) The Jaccard score on test data at each epoch with different TSs.

Concerning the above limitations, potential solutions might be carried out from two perspectives. First, more flexible HMC modules can be designed to handle more complex hierarchical label systems in HMC data sets. Second, more advanced curriculum learning technology can be designed to automatically select hyperparameters more effectively and efficiently.

8. Conclusion

Predicting medications with longitudinal EHRs for patients (especially for multi-morbidity patients) in ICUs can provide a heuristic guideline for clinicians and improve the efficiency of clinical decision-making. In this paper, we propose a curriculum learning enhanced hierarchical multi-label classification framework for medication recommendation (CEHMR). CEHMR mainly includes a well-designed multi-label classification network (HMR) and a bootstrap-based training algorithm (SGCL). HMR can capture the hierarchical dependency among multi-level medication labels in the medication category hierarchy such as ATC. SGCL can alleviate the negative impact caused by the diversity of training examples and improve overall performance. Extensive experiments on the MIMIC-III database demonstrate that our framework achieves superior performance compared with state-of-the-art baselines. The ablation study and analysis further illustrate the effectiveness of each CEHMR component. Besides, because the proposed CEHMR can predict high-level medication codes more accurately, it may better assist clinicians with various purposes. CEHMR also has some limitations such as capacity for complex hierarchical label systems and manual labor for hyperparameter selection. In our future works, we plan to focus on exploration of more flexible HMC designs and automatic curriculum learning technologies.

Declaration of competing interest

We confirm that this manuscript is entirely original, has not been copyrighted, published, submitted, or accepted for publication elsewhere. We have no conflicts of interest to disclose. I look forward to learning your response to our submission.

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References

[1] Spaulding EM, Marvel FA, Lee MA, Yang WE, Demo R, Wang J, Xun H, Shah L, Weng D, Fashanu OE, et al. Corrie health digital platform for self-management in secondary prevention after acute myocardial infarction: Micore rationale and design. Circul: Cardiovasc Qual Outcomes 2019;12(5):e005509. http://dx.doi.org/10.1161/CIRCOUTCOMES.119.005509.

[2] Choi E, Bahadori MT, Sun J, Kulas J, Schuetz A, Stewart W. Retain: An interpretable predictive model for healthcare using reverse time attention mechanism. Adv Neural Inf Process Syst 2016;29. http://dx.doi.org/10.48550/arXiv.1608.05745.

[3] Shang J, Xiao C, Ma T, Li H, Sun J. Gamenet: Graph augmented memory networks for recommending medication combination. In: Proceedings of the AAAI conference on artificial intelligence, Vol. 33. 2019, p. 1126–33. http://dx.doi.org/10.1609/aaai.v33i01.33011126.

[4] Shang J, Ma T, Xiao C, Sun J. Pre-training of graph augmented transformers for medication recommendation. In: Proceedings of the twenty-eighth international joint conference on artificial intelligence, IJCAI-19. International Joint Conferences on Artificial Intelligence Organization; 2019, p. 5953–9. http://dx.doi.org/10.24963/ijcai.2019/825.

[5] He Y, Wang C, Li N, Zeng Z. Attention and memory-augmented networks for dual-view sequential learning. In: Proceedings of the 26th ACM SIGKDD international conference on knowledge discovery & data mining. 2020, p. 125–34. http://dx.doi.org/10.1145/3394486.3403055.

[6] Zhang Y, Chen R, Tang J, Stewart WF, Sun J. LEAP: learning to prescribe effective and safe treatment combinations for multimorbidity. In: Proceedings of the 23rd ACM SIGKDD international conference on knowledge discovery and data mining. 2017, p. 1315–24. http://dx.doi.org/10.1145/3097983.3098109.

[7] Wang S, Ren P, Chen Z, Ren Z, Ma J, de Rijke M. Order-free medicine combination prediction with graph convolutional reinforcement learning. In: Proceedings of the 28th ACM international conference on information and knowledge management. 2019, p. 1623–32. http://dx.doi.org/10.1145/3357384.3357965.

[8] Le H, Tran T, Venkatesh S. Dual memory neural computer for asynchronous two-view sequential learning. In: Proceedings of the 24th ACM SIGKDD international conference on knowledge discovery & data mining. 2018, http://dx.doi.org/10.1145/3219819.3219981.

[9] Wang H, Wu Y, Gao C, Deng Y, Zhang F, Huang J, Liu J. Medication combination prediction using temporal attention mechanism and simple graph convolution. IEEE J Biomed Health Inf 2021;25(10):3995–4004. http://dx.doi.org/10.1109/JBHI.2021.3082548.

[10] Futoma J, Morris J, Lucas J. A comparison of models for predicting early hospital readmissions. J Biomed Inform 2015;56:229–38. http://dx.doi.org/10.1016/j.jbi.2015.05.016.

[11] Ashfaq A, Sant'Anna A, Lingman M, Nowaczyk S. Readmission prediction using deep learning on electronic health records. J Biomed Inform 2019;97:103256.

[12] Wang S, Li X, Chang X, Yao L, Sheng QZ, Long G. Learning multiple diagnosis codes for ICU patients with local disease correlation mining. ACM Trans Knowl Discov Data (TKDD) 2017;11(3):1–21. http://dx.doi.org/10.1016/j.jbi.2019.103256.

[13] Gao J, Wang X, Wang Y, Yang Z, Gao J, Wang J, Tang W, Xie X. Camp: Co-attention memory networks for diagnosis prediction in healthcare. In: 2019 IEEE international conference on data mining (ICDM). IEEE; 2019, p. 1036–41. http://dx.doi.org/10.1109/ICDM.2019.00120.

[14] Zhang X, Qian B, Li Y, Yin C, Wang X, Zheng Q. KnowRisk: an interpretable knowledge-guided model for disease risk prediction. In: 2019 IEEE international conference on data mining (ICDM). IEEE; 2019, p. 1492–7. http://dx.doi.org/10.1109/ICDM.2019.00196.

[15] Chen C, Liang J, Ma F, Glass L, Sun J, Xiao C. Unite: Uncertainty-based health risk prediction leveraging multi-sourced data. In: Proceedings of the web conference 2021. 2021, p. 217–26. http://dx.doi.org/10.1145/3442381.3450087.

[16] Zhang M-L, Zhou Z-H. ML-KNN: A lazy learning approach to multi-label learning. Pattern Recognit 2007;40(7):2038–48. http://dx.doi.org/10.1016/j.patcog.2006.12.019.

[17] Liu W, Tsang IW, Müller K-R. An easy-to-hard learning paradigm for multiple classes and multiple labels. J Mach Learn Res 2017;18(94):1–38.

[18] Wang X, Chen Y, Zhu W. A survey on curriculum learning. IEEE Trans Pattern Anal Mach Intell 2021. http://dx.doi.org/10.1109/TPAMI.2021.3069908.
[19] Wei J, Suriawinata A, Ren B, Liu X, Lisovsky M, Vaickus L, Brown C, Baker M, Nasir-Moin M, Tomita N, et al. Learn like a pathologist: curriculum learning by annotator agreement for histopathology image classification. In: Proceedings of the IEEE/CVF winter conference on applications of computer vision. 2021, p. 2473–83.
[20] Johnson AE, Pollard TJ, Shen L, Lehman L-wH, Feng M, Ghassemi M, Moody B, Szolovits P, Anthony Celi L, Mark RG. MIMIC-III, a freely accessible critical care database. Sci Data 2016;3(1):1–9.
[21] Rousu J, Saunders C, Szedmak S, Shawe-Taylor J. Kernel-based learning of hierarchical multilabel classification models. J Mach Learn Res 2006;7:1601–26.
[22] Cesa-bianchi N, Gentile C, Tironi A, Zaniboni L. Incremental algorithms for hierarchical classification. In: Saul L, Weiss Y, Bottou L, editors. Advances in neural information processing systems, Vol. 17. MIT Press; 2004.
[23] Mayne A, Perry R. Hierarchically classifying documents with multiple labels. In: 2009 IEEE symposium on computational intelligence and data mining. IEEE; 2009, p. 133–9. http://dx.doi.org/10.1109/CIDM.2009.4938640.
[24] Lewis DD, Yang Y, Russell-Rose T, Li F. Rcv1: A new benchmark collection for text categorization research. J Mach Learn Res 2004;5(Apr):361–97.
[25] Dimitrovski I, Kocev D, Loskovska S, Džeroski S. Hierarchical annotation of medical images. Pattern Recognit 2011;44(10–11):2436–49. http://dx.doi.org/10.1016/j.patcog.2011.03.026.
[26] Barros RC, Cerri R, Freitas AA, de Carvalho AC. Probabilistic clustering for hierarchical multi-label classification of protein functions. In: Joint European conference on machine learning and knowledge discovery in databases. Springer; 2013, p. 385–400. http://dx.doi.org/10.1007/978-3-642-40991-2_25.
[27] Cerri R, Barros RC, de Carvalho AC. Hierarchical classification of gene ontology-based protein functions with neural networks. In: 2015 international joint conference on neural networks (IJCNN). IEEE; 2015, p. 1–8. http://dx.doi.org/10.1109/IJCNN.2015.7280474.
[28] Triguero I, Vens C. Labelling strategies for hierarchical multi-label classification techniques. Pattern Recognit 2016;56:170–83. http://dx.doi.org/10.1016/j.patcog.2016.02.017.
[29] Silla CN, Freitas AA. A survey of hierarchical classification across different application domains. Data Min Knowl Discov 2011;22(1):31–72. http://dx.doi.org/10.1007/s10618-010-0175-9.
[30] Costa EP, Lorena AC, Carvalho AC, Freitas AA, Holden N. Comparing several approaches for hierarchical classification of proteins with decision trees. In: Brazilian symposium on bioinformatics. Springer; 2007, p. 126–37.
[31] Kiritchenko S, Matwin S, Famili AF, et al. Hierarchical text categorization as a tool of associating genes with gene ontology codes. In: European workshop on data mining and text mining in bioinformatics. 2004, p. 30–4.
[32] Cerri R, Barros RC, de Carvalho AC. A genetic algorithm for hierarchical multi-label classification. In: Proceedings of the 27th annual ACM symposium on applied computing. 2012, p. 250–5. http://dx.doi.org/10.1145/2245276.2245325.
[33] Wehrmann J, Cerri R, Barros R. Hierarchical multi-label classification networks. In: International conference on machine learning. PMLR; 2018, p. 5075–84.
[34] Wei Y, Liang X, Chen Y, Shen X, Cheng M-M, Feng J, Zhao Y, Yan S. Stc: A simple to complex framework for weakly-supervised semantic segmentation. IEEE Trans Pattern Anal Mach Intell 2016;39(11):2314–20.
[35] Bengio Y, Louradour J, Collobert R, Weston J. Curriculum learning. In: Proceedings of the 26th annual international conference on machine learning. 2009, p. 41–8. http://dx.doi.org/10.1145/1553374.1553380.
[36] Guo S, Huang W, Zhang H, Zhuang C, Dong D, Scott MR, Huang D. Curriculumnet: Weakly supervised learning from large-scale web images. In: Proceedings of the European conference on computer vision (ECCV). 2018, p. 135–50.
[37] El-Bouri R, Eyre D, Watkinson P, Zhu T, Clifton D. Student-teacher curriculum learning via reinforcement learning: predicting hospital inpatient admission location. In: International conference on machine learning. PMLR; 2020, p. 2848–57.
[38] Platanios EA, Stretcu O, Neubig G, Póczos B, Mitchell T. Competence-based curriculum learning for neural machine translation. In: Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers). 2019, p. 1162–72.
[39] Tsvetkov Y, Faruqui M, Ling W, MacWhinney B, Dyer C. Learning the curriculum with Bayesian optimization for task-specific word representation learning. In: Proceedings of the 54th annual meeting of the association for computational linguistics. Association for Computational Linguistics; 2016, p. 130–9. http://dx.doi.org/10.18653/v1/P16-1013.
[40] Liu C, He S, Liu K, Zhao J. Curriculum learning for natural answer generation. In: Proceedings of the twenty-seventh international joint conference on artificial intelligence, IJCAI-18. International Joint Conferences on Artificial Intelligence Organization; 2018, p. 4223–9. http://dx.doi.org/10.24963/ijcai.2018/587.
[41] Yuan X, He P, Zhu Q, Li X. Adversarial examples: Attacks and defenses for deep learning. IEEE Trans Neural Netw Learn Syst 2019;30(9):2805–24. http://dx.doi.org/10.1109/TNNLS.2018.2886017.
[42] Kumar M, Packer B, Koller D. Self-paced learning for latent variable models. In: Lafferty J, Williams C, Shawe-Taylor J, Zemel R, Culotta A, editors. Advances in neural information processing systems, Vol. 23. Curran Associates, Inc.; 2010, p. 1189–97.
[43] Weinshall D, Cohen G, Amir D. Curriculum learning by transfer learning: Theory and experiments with deep networks. In: Dy J, Krause A, editors. Proceedings of the 35th international conference on machine learning. Proceedings of machine learning research, Vol. 80, PMLR; 2018, p. 5238–46.
[44] Hacohen G, Weinshall D. On the power of curriculum learning in training deep networks. In: Chaudhuri K, Salakhutdinov R, editors. Proceedings of the 36th international conference on machine learning, ICML 2019, 9-15 June 2019, Long Beach, California, USA. Proceedings of machine learning research, Vol. 97, PMLR; 2019, p. 2535–44.
[45] Graves A, Bellemare MG, Menick J, Munos R, Kavukcuoglu K. Automated curriculum learning for neural networks. In: Precup D, Teh YW, editors. Proceedings of the 34th international conference on machine learning. Proceedings of machine learning research, Vol. 70, PMLR; 2017, p. 1311–20.
[46] Matiisen T, Oliver A, Cohen T, Schulman J. Teacher–student curriculum learning. IEEE Trans Neural Netw Learn Syst 2019;31(9):3732–40.
[47] Tatonetti NP, Ye PP, Daneshjou R, Altman RB. Data-driven prediction of drug effects and interactions. Sci Transl Med 2012;4(125):125ra31.
[48] Kingma DP, Ba J. Adam: A method for stochastic optimization. 2014, arXiv preprint arXiv:1412.6980.